

REGULAR ARTICLE

Ex-post evaluation of fishery management policies on wild fisheries production in northern Cabo Verde: An example of mackerel scad (*Decapterus macarellus*, Carangidae)

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Abstract

Government policies for marine fisheries have been implemented in Cabo Verde since its independence in 1975, with the aim to prevent overexploitation of wild fish species and promote sustainable fishing practices. Nonetheless, only minor amendments have been made to the legal harvesting size, considering biological sciences. This study, therefore, adopted a transdisciplinary approach to assess the political, ecological, and social dimensions of current fishery policy interventions applicable to the commercially valuable pelagic species *Decapterus macarellus* (mackerel scad). An ex-post analysis of relevant fishery management policies targeting *D. macarellus* was conducted. This was supplemented by strengths, weaknesses, opportunities, and threats (SWOT) analysis conducted by key fisheries stakeholders. Stocks assessment was conducted on catch data before (2003–2007) and after (2017–2021) the policy interventions. This was followed by a survey of 175 fishery sector actors to understand their perception of the policies, compliance challenges, and recommendations for reforms. Results showed that although the mackerel scad landing size comparably increased, landing catches were on the decline. Most fisheries stakeholders are aware of the policies implemented and acknowledge their favorable developmental outcomes. However, certain gaps exist in the national marine fisheries policies. For example, there is an absence of follow-up research on implemented fishing policies and a lack of monitoring data on the ecology and distribution of the mackerel scad, which hinders our understanding of the exact causes of the reported continual decrease in catches. There is, therefore, a need for regular monitoring of the environmental health of coastal and marine habitats to inform prioritization and/or reformulation of policy intervention measures to achieve intended conservation outcomes.

KEYWORDS

biological rest period, fish, fishing policies, landing size, overexploitation

1 | INTRODUCTION

Small island developing states (SIDS) are particularly vulnerable to the impacts of climate change (Monnereau et al., 2021; Robinson et al., 2022). Although SIDS are a diverse group of countries, one shared feature is their extremely large economic exclusive zone (EEZ) to land mass ratio, highlighting the significance of their marine and coastal resources (Monnereau et al., 2021). Fish and other seafood are essential sources of protein and livelihoods for many island communities (Hay, 2013). Fish provides at least 15% of the average per capita intake of animal protein for 4.5 billion people (Béné et al., 2015; Canyon et al., 2021). Globally, per capita fish consumption has increased approximately twice as fast as the annual global population growth, from 9.0 kg in 1961 to 20.5 kg in 2018, resulting in fish being the most consumed animal protein in 2018 (Food and Agriculture Organization [FAO], 2020). In Africa, fisheries play a significant role in supporting household livelihoods (Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services [IPBES], 2018), providing a source of income for 8 million fishers (Belhabib et al., 2015; Teh et al., 2013), and contributing up to a total of \$20 billion to the national gross domestic product (GDP) (Belhabib et al., 2016). Among the African regions, West Africa has the highest fisheries landing catch (Sambe et al., 2011), estimated at 3.5 million tons annually (FAO, 2020).

Cabo Verde, a developing country in the East Atlantic Ocean, is classified under the Atlantic, Indian Ocean, Mediterranean Sea, and South China Sea (AIMS) category of SIDS, together with eight other island states (Bahrain, Comoros, Guinea-Bissau, Maldives, Mauritius, Sao Tome and Principe, Seychelles, and Singapore). The archipelago is characterized by a narrow insular shelf of volcanic origin, unpredictable patterns of rainfall, harsh climatic and oceanographic conditions, and lacks upwelling systems (Direção Nacional do Ambiente [DNA], 2015a; INDP, 2011b). Its marine resources are characterized by high diversity but low abundance (PANA II, 2004). Compared to adjacent upwelling systems near the West African coasts, seawater surrounding the Cabo Verde islands is oligotrophic (World Bank, 2008). Fisheries contribute between 2% and 10% to the national GDP (DNA, 2015a). The statistical data of the National Fisheries institute, IMar, formerly known as Instituto Nacional para o Desenvolvimento das Pescas (INDP), estimate that in 2018, fish landing catch was 26,588 t with tuna, small pelagic, and demersal fish dominating the species composition. In Cabo Verde, the primary source of animal protein for human consumption is fish (González and Tariche, 2009; Segundo Plano de Ação Nacional para o Ambiente [PANA II], 2004), comprising an estimated 70% of the total (Costa et al., 2020). Fish consumption per capita increased from 19 kg in 1998 to 26.5 kg in 2004 (PANA II, 2004; National Institute for Fisheries Development [INDP], 2012), exacerbating the pressure on marine resources (Tariche et al., 2011). Fish is an important source of essential amino acids, micronutrients, and long-chain polyunsaturated fatty acids (LC-PUFAs) and is therefore an attractive solution to malnutrition and food insecurity, particularly in lower-income and food-insecure countries (Béné et al., 2015; Kawarazuka & Béné, 2011).

Furthermore, fisheries exhibit a lower carbon production footprint in contrast to other animal production processes (Béné et al., 2015). Fisheries also provide other benefits such as recreation, direct and indirect employment opportunities, (international) trade, and overall socioeconomic well-being (FAO, 1995; FAO, 2020). Therefore, there is a need to ensure continuous wild fish production by promoting sustainable fishery management policies and practices (Hilborn et al., 2020).

The fisheries management policies of Cabo Verde, as set out in the decree-law 3/2005 (República de Cabo Verde, 2005), have been enacted to mitigate marine fisheries overexploitation and promote sustainable fishery (PANA II, 2004). The policy measures aimed to reduce fishing pressures on important commercial species, including *Decapterus macarellus*, Cuvier & Valenciennes, 1833 (locally known as “cavala preta,” called mackerel scad in this study), involve imposing quotas on the minimum landing size (MLS) and restricting fishing efforts by implementing no-fishing seasons or biological rest periods (BRPs). Nevertheless, both regulations were developed based on landing catches and biological studies that failed to consider the reproductive biology, specifically, the microscopic gonad characteristics (Costa et al., 2020).

Mackerel scad, a species of fish of the Carangidae family, is the most important commercial small pelagic fish consumed in Cabo Verde (Almada, 1997b) with high nutritional value and affordability (INDP, 2007). Since 2008, harvesting of this fish has been regulated by fishery policies (República de Cabo Verde, 2009). Among the key measures outlined in the fisheries management plan are (i) delimitation of fishing zones, (ii) the prohibition of harmful fishing practices, (iii) the establishment of minimum catch sizes of target species, and (iv) implementing closed seasons for endangered species (República de Cabo Verde, 2021). Furthermore, fisheries policy reforms have been insufficient, and currently mackerel scad is to be harvested by semi-industrial purse seine vessels over 8 m long, as well as small-scale vessels of 8–10 m long equipped with an outboard motor. The minimum landing size (MLS) is 20 cm furcal length (FL) and 6 cm FL for its usage as bait in the tuna pole and line fishery, whereas the BRP is from July 15 to September 14. This was, however, suspended in 2020 and 2021 due to the COVID-19 pandemic, as government subsidies were not provided for actors in the fishery sector (República de Cabo Verde, 2021) (Table 1). Following this policy intervention (i.e., the MLS and BRP), there was a short-term improvement in both the quantity and quality of mackerel scad catches, resulting in higher income for fishers (DNA, 2015b). Official landing catch data from 1989 to 2015 reveal that mackerel scad constituted nearly 40% of the total catches of Cabo Verde, with the highest landing catches occurring in 1997 and 1998 (da Luz & Vieira, 2020). However, there has been a substantial reduction in the catch since then, and in 2015, it represented only 6.6% of the total catch (Vieira, 2018). Like mackerel scad, wild populations of other pelagic and demersal species, such as the pink lobster, *Palinurus charlestoni* (Forest & Postel, 1964), and crustaceans, are also experiencing declining numbers (DeAlteris, 2012; DNA, 2015b).

Despite their importance, policy interventions and resulting developments within the marine fisheries sector of Cabo Verde are

TABLE 1 Summary of the historical evolution of fisheries policies for *Decapterus macarellus* implemented in Cabo Verde between 2008 and 2021.

Resolution no. 11/2007	Resolution no. 10/2009	Resolution no. 56/2014	Resolution no. 1/2016	Resolution no. 29/2016	Resolution no. 8/2018	Resolution no. 93/2020	Resolution no. 63/2021
2007–2008	2009–2010	2014–2015	2016	2016–2017	2018	2020	2021
Authorize fishing by industrial purse seiner, usually multipurpose, depending on fish species availability.							
Fishing is reserved for national vessels.							
Cautious expansion of fishing efforts through fishing licenses control.							
Allowed for purse seiner bigger than 6.5 m in length.		Allowed for purse seiner bigger than 8 m in length.				Allowed for purse seiner between 8 and 10 m, with outboard motor.	
Establishment of biological rest period between August 1 and September 30, starting from January 2008.				Establishment of biological rest period, between July 15 and September 14.		Exceptional suspension of biological rest period, (due to COVID-19 pandemic)	
-	Margin of tolerance allowed 10% of the total weight of the catch in the case of by-catches of mackerel scad in the purse seine fishery.						
Establishment of minimum landing size, to be marketed, of 18 cm furcal length, starting in January 2008.				Establishment of minimum landing size, to be marketed, of 20 cm furcal length.			
-	Ban on the marketing of mackerel scad caught on the previous terms, except when used as live bait. The minimum size for bait is kept at 6 cm (small-scale fishing).						
-	Authorization of semi-industrial seine fishing within 3 nautical miles exclusively for the capture of small pelagic, except in all bays of the archipelago.						

not subjected to rigorous evaluation, primarily, due to insufficient funds (República de Cabo Verde, 2014). Therefore, there is a need for an in-depth analysis of the effectiveness of the national fisheries laws and policies in terms of sustainably managing wild species and restoring stock levels. Therefore, this study aimed to (1) evaluate the effectiveness of marine fisheries management policy, specifically MLS quotas and BRP in the north of Cabo Verde targeting the restoration of mackerel scad stock; (2) assess the biometric indices and population parameters of commercial catches 5 consecutive years prior to the fishery policy intervention in 2008 (i.e., from 2003 to 2007) and after the policy interventions (i.e., from 2017 to 2021); and (3) assess the perceptions of state and non-state actors in the fisheries sector regarding the policies and challenges surrounding compliance with a view to proffer recommendations for policy reforms.

2 | MATERIALS AND METHODS

2.1 | Description of the study area

Cabo Verde is located 600 km off the West African coast, geographically between 16.5388° N and 23.0418° W in Northwest Africa. The country has a surface area of 4033 km², a shoreline of about 2000 km (Martins & Knútsson, 2002), and an EEZ of 789,400 km² (World Bank, 2008). In 2021, the country reported a population of 587,925 (World Bank, 2021). The country comprises 10 islands of volcanic origin, arising from depths of at least 3000 m (World Bank, 2008). The islands are divided into the Barlavento (windward) islands (Santo Antão, São Vicente, Santa Luzia, São Nicolau, Sal, and Boavista) and

the Sotavento (leeward) islands (Maio, Santiago, Fogo, and Brava), from north to south.

Cabo Verde lies in the dry tropical Sahelian zone characterized by an arid to semi-arid climate, influenced by the Azores anticyclone, the Intertropical Convergence Zone (ITCZ), and the movements of Mid-Atlantic air masses (Duarte & Romeiras, 2009). The country has experienced periodic drought throughout its history and is currently going through a particularly severe 20-year period of abnormal rains (Monteiro et al., 2020). The annual rainfall season occurs mainly from August to October, with a maximum precipitation amount of 122.2 mm (mean ± SD of 148.5 ± 15.9 mm) (Hijmans et al., 2005), and the evaporation exceeds the precipitation (Duarte & Romeiras, 2009). The mean annual temperature ranges between 22.2 and 23.8°C (mean ± SD of 23.1 ± 0.5°C), but during the warmest months it can reach 26.7 to 28.1°C (±0.4°C), and during the coolest months, it drops to 18.0 to 19.4°C (±0.4°C) (Hijmans et al., 2005). The ocean and the northeastern trade winds, which bring moderate humidity year-round, help moderate the climate (Hijmans et al., 2005). The northern part of the country, influenced by the Canary current, has a colder sea surface temperature (SST) than the southern part, with an average of 24°C.

2.2 | Analyses of marine fisheries management policy documents in Cabo Verde

This study adopted a transdisciplinary approach, adopting methodologies from ex-post policy analysis (Organisation for Economic Co-operation and Development [OECD], 2016), stock assessment (Cadima, 2000; Sparre & Venema, 1997), and sociopolitical studies

TABLE 2 Strengths, weaknesses, opportunities, and threats analysis for mackerel scad management policies implemented in Cabo Verde derived from an expert group that included fishers from fishing communities of São Vicente Island.

Fisheries experts	Strengths	Weaknesses	Opportunities	Threats
Joint opinion		-Weak enforcement -Absence of monitoring framework		-Lack of ex-post evaluation -Insufficient landing port surveillance efforts
Fishers	Decrease in fishing mortality during the closed season	-Reduced income from fishing during the closed season -Low involvement of fishers during the management decision-making processes	-Studies to understand the disappearance of mackerel scad in commercial catches -Improve and increase fishing efforts (e.g., provision of bigger and more sophisticated vessels)	-Overexploitation of apex predators (tuna species) is responsible for the offshore migration of the mackerel scad
Fishery management institutions	Existence of fisheries management policies -Reduction in fishing pressure on the mackerel scad, leading to recovery in stock and decrease in fish mortality -Protection of spawning grounds of the mackerel scad	MLS and BRP are misaligned with the specificity of the species -Social exclusion in the policy-making process	-Create enabling environment for scientific research to inform policy reforms	-Legal harvesting size less than the size at first maturity will compromise conservation efforts -The negative impacts of abiotic factors, such as climate change and ocean surface warming, imply the migration of species to more suitable habitats.
Academia and NGOs	-Promotion of sustainable fisheries production -Exploitation of underexploited and underutilized species -Increase in minimum landing size	-Weak dissemination/lack of awareness of existing laws -Lack of social inclusion/experts' opinion in policies formulation/monitoring -Limited studies on wild stock assessment	-Development and/or strengthening of partnerships with NGOs and fishing communities -Raise awareness of fisheries policies and management plans -Integrate and value knowledge of indigenous and local fishers -Develop strategies for wild stock monitoring and policy (re)formulation -Stakeholder involvement in fisheries decision-making processes	-Lack of awareness of fisheries laws and policies -Lack of funds to hire scientists for scientific assessments -Absence of alternative income-generating activities for fisheries-dependent communities during closed seasons

Abbreviations: BRP, biological rest period; MLS, minimum landing size; NGO, nongovernmental organization.

(Kelemen et al., 2021). The national marine fishery management policies and official bulletins of the associated Fisheries resources management plan, published since the enactment of the laws in 2008, were assessed. Policy document analysis was complemented with an expert consultation roundtable held on October 15, 2021, which involved 20 key fisheries stakeholders, including fishers, fisheries management institutions, academia, and nongovernmental organizations (NGOs). The goal of this multi-stakeholder consultation was to conduct a SWOT (strengths, weaknesses, opportunities, and threats) analysis of the fisheries management policies, focusing on the BRP and MLS targeted at mackerel scad stock. The participants were split into groups based on their roles within the fishery sector, and each group carried out the SWOT analysis based on their perspectives and understanding of the policies. NGOs and academia representatives were combined in a single group. The results from each group activity are presented in Table 2.

2.3 | Assessment of biometric indices and population parameters of mackerel scad from commercial fish landings

Biometric data on mackerel fish catch were obtained from the national fish database of IMar. The data are publicly accessible at DOI:10.5281/zenodo.10062019. Samples of mackerel scad were collected from marine fish landings at the fishing port of Cova d'Inglesa. As reported by the fishers, mackerel scad was harvested on multispecific trips with the same gear, mostly from waters around Ilhéu Branco, Ilhéu Raso, Sal, Santa Luzia, Santo Antão, São Nicolau, and São Vicente, areas of the Barlavento islands, northern Cabo Verde. The last research expedition to estimate the abundance of pelagic fish population in 1997 indicated an abundance of mackerel scad of 2700 t/year (Almada, 1997a). To distinguish between pre- and post-policy interventions, catch data from 2003 to 2007 were aggregated to

represent the 5-year pre-policy intervention phase (2003–2007). Similarly, data from 2017 to 2021 were aggregated to represent the post-policy intervention phase (2017–2021). Biometric indices, including length frequency distribution and growth in length, were computed to determine the effectiveness of imposing quotas on MLS using the Von Bertalanffy (1938) model adapted by Beverton and Holt (1956):

$$L(t) = L_{\infty} \times \left[1 - e^{-K(t-t_0)} \right] \quad (1)$$

where growth parameters (L_{∞} and K) were estimated using the FAO-ICLARM Stock Assessment Tools (FiSAT II software), and $t = t_0$ was estimated using Pauly model (Pauly & Moreau, 1997):

$$\text{Log}(-t_0) = -0.392 - 0.275 \text{Log } L_{\infty} - 1.0381 K. \quad (2)$$

To assess the effectiveness of the BRP, sex was determined through the macroscopic examination of the gonads. Maturity stages were assigned to the gonads using the national gonad maturity guide adapted by IMar. Reproduction timing was evaluated by calculating the gonadosomatic index (GSI) following the equation of King (1995):

$$\text{GSI} = 100 \times \left(\frac{\text{GW}}{\text{TW}} \right) \quad (3)$$

where GW is gonad weight, and TW is total tissue weight. Size at first sexual maturation is commonly determined using the Haddon logistic model, L_{50} method, which establishes the length (l) at which 50% of the sampled specimens are mature (P) (Haddon, 2001):

$$P = \frac{1}{1 + e^{\frac{-(l-L_{50})}{\delta}}} \quad (4)$$

Selected fish life-history parameters were subjected to Kruskal–Wallis test to check for significant differences between specimens in both study periods using R (R Core team, 2013). Stock assessments were computed using spreadsheet and FiSAT II software (Gayanilo & Sparre, 2005) based on fish stock assessment methods proposed by Sparre and Venema (1997), Cadima (2000), Tariche and Martins (2006), and Santos (2018).

2.4 | Social survey of fisheries stakeholders

The survey was conducted among diverse groups of fisheries stakeholders to assess their awareness, knowledge, and perceptions of the implemented fishery policies for mackerel scad. The ethical approval for this survey was given by IMar, and prior to the survey, all participants granted their consent. Anonymity was ensured during data collection and processing. Structured questionnaires were distributed via online surveys to stakeholders from fishery management institutions, NGOs, academia, and civil society (consumers), and face-to-face interviews were conducted with fisheries actors at the landing ports and

fish markets in São Pedro and Salamansa villages in São Vicente, who lacked access to digital technology. Assessment of knowledge on fisheries management policies was based on both subjective self-assessment by respondents and objective measures, which included the percentage of correct responses to a set of questions, measured using Pearson's correlation coefficient. Data were analysed using descriptive statistics with STATA version 15 (Statacorp LLC, USA), as well as probit regression to examine the correlation between sociodemographic characteristics and policy awareness. This descriptive analysis did not control for various sources of endogeneity due to data limitations, making causal inferences impossible. Thus, our findings only demonstrate associations. To test whether there is a correlation between sociodemographic characteristics and objectively measured knowledge of the policy interventions, a probit model was estimated in which a dependent variable, that is, a dummy equal to one if a respondent correctly answered questions relating to either the MLS or the BRP, was regressed on several variables, including gender, age, education, and years of fishing experience.

3 | RESULTS

3.1 | Marine fishery management policies in Cabo Verde: An overview

Cabo Verde has been implementing policies to manage its fisheries resources since colonial times, as stipulated in decree number 7/1879. The adoption of sustainable policies increased after the ratification of the United Nations Convention on the Law of the Sea (UNCLOS) (law number: 17/III/87). Consequently, the fundamental principles for sustainable exploitation of fisheries resources were established in 1987 by decree law number 17/87. Additionally, decree number 97/87 implemented measures to safeguard, conserve, and regulate the exploitation of the ocean and marine resources (INDP, 2011b). The management of Cabo Verdean marine fisheries is reliant on three distinct institutional pillars: (i) the Fisheries Research Institute (IMar), which serves as the scientific advisory body in charge of developing the knowledge base necessary for fisheries management; (ii) the National Directorate for Fisheries and Aquaculture, which represents the fisheries administration and utilizes evidence-based recommendations to formulate and implement fisheries policies; and (iii) the General Directorate for Fisheries Inspection, Maritime and Portuary Institute, Maritime Police and National Police, the inspection body responsible for enforcing compliance (INDP, 2004; PANA II, 2004). The fisheries management strategies are documented in the Fisheries Resources Management Plan (INDP, 2011a; PANA II, 2004). The primary objective is to promote the development of the fisheries sector in alignment with the National Strategic Plan for Sustainable Development (PEDS, 2017–2021) and the framework of the Blue Growth Charter. Additionally, it aims to ensure sustainable socioeconomic growth and the development of value-added fish and fishery products, and thereby contribute to food security (República de Cabo Verde, 2020).

3.1.1 | SWOT analysis of mackerel scad management policies: Key stakeholders' perspectives

The SWOT analysis showed that all stakeholder groups agreed that the fisheries policy has been beneficial for the mackerel scad but stressed the need for a transparent and participatory policy (re)formulation process (Table 2). Fishers, while broadly supportive of the policy instrument, expressed dissatisfaction with the timing of the BRP. They suggested that the timing of the BRP should be adjusted as it is to blame for decline in catches. They opine that mackerel scad spawn all year round and that the current BRP is the time when mackerel scad is most abundant, forms large schools, and migrates to other areas. As a result, mackerel scad is being harvested further away from the coast, where semi-industrial boats with larger nets are used during the open season. The fisheries management institution believes that the causes of the decline are largely unknown but may be due to other factors, such as climate change and population growth, and that there is a need to consider the social impact of recommended policies, which have so far been based on biological science.

The fishers expressed the lack of government support during periods of low mackerel scad catches. This is particularly important given that fishing is their only economic activity of which mackerel scad is an important part. In response, the fishery management institution urged the fishers to exploit underexploited seafood, such as crabs, as well as big-eyed tuna (*Thunnus albacares* Bonnaterre, 1788), which is open to fishing during the BRP of mackerel scad, and more profitable because they are bigger than mackerel scad and mostly sold to seafood processing industries (República de Cabo Verde, 2021).

A second reason put forward by fishers for the decline in mackerel scad is the increase in foreign industrial fleets that harvest large apex predatory pelagic fish, such as tuna, known predators of mackerel scad. According to the fishers, mackerel scad migrates to the coast and upper waters to escape predation. This is, however, not the case with the collapse of the apex predators, as they migrate elsewhere, highlighting the trophic impacts of fishery policies. This observation has been confirmed by previous studies (McNaughton, 2008). This claim by fishers may be valid, as a comparison of landing catches of high-trophic level species such as tuna, other pelagic species, and mackerel scad from 1992 to 2020 shows that landings of the yellow-fin tuna were increasing, whereas all other catches, including mackerel scad, were decreasing.

3.2 | Biometric indices of mackerel scad prior to, and after policy interventions

Results from the data analysis of 7512 specimens of mackerel scad distributed in two groups, before (years 2003–2007; $n = 4424$ specimens) and after (years 2017–2021; $n = 3039$ specimens), showed that the mean ($\pm$ SD) fork length of harvested species in 2003/2007 was 26.73 ± 2.81 cm, significantly lower (p -value $< 2.2e-16$) than those harvested between 2017 and 2021 with a mean ($\pm$ SD) of 29.07 ± 3.20 cm. The length frequency was between 16 and 35 cm before

the policy intervention and between 18 and 45 cm after the policy intervention, as shown by the shift and flattening of the normality curve (Figure 1). The mean ($\pm$ SD) body weight was 255.7 ± 70 g for 2003–2007 and 312.8 ± 113.5 g for 2017–2021, which represents a highly significant (p -value $< 2.2e-16$) increase of $\sim 22\%$ in the body weight of its landing catches. Consequently, growth parameters were relatively higher in the second period ($L_{\infty} = 46$ cm FL; $K = 0.47 \text{ year}^{-1}$) compared to the first period ($L_{\infty} = 41.48$; $K = 0.39 \text{ year}^{-1}$) (Figure 2). The mean ($\pm$ SD) gonad weight was 7.41 ± 6.7 g in 2003–2007, significantly lower (p -value $< 1.723e-15$) than in 2017–2021 with an observed mean ($\pm$ SD) of 9.15 ± 11.81 g. However, the percentage of sexually matured fish was significantly higher (p -value $< 2.2e-16$) in 2003–2007 (76%) compared to 2017–2021 (57%). The results of the GSI analysis indicate that it is possible to find mackerel scad reproducing throughout the year, but with a higher representation between February and October (Figure 3). Regarding BRP, the Haddon logistic model showed that the length at first maturity, L_{50} , increased from 24.2 to 28.2 cm for the second period of analysis (Figure 4).

3.3 | Stakeholders' perception of mackerel scad fishing management policies

In total, 175 randomly selected respondents were surveyed, of those 100 were done online and 75 were face-to-face interviews. Table 3 shows their sociodemographic characteristics. There are significant differences between stakeholders in terms of gender, educational level, and years of experience in fisheries. For example, respondents involved in fisheries management, academia, and NGOs have the highest level of education, whereas fishers are the group with the highest number of years of work experience (Figure 5).

In the survey, respondents were asked to assess, in a subjective manner, their knowledge of the mackerel scad fisheries-related policy interventions implemented in Cabo Verde, that is, the MLS (i.e., *Do you know the minimum landing size for mackerel scad?*) and the BRP (i.e., *Do you know the biological rest period for mackerel scad?*). This was followed by an objective assessment based on questions (i.e., *What is the minimum landing size for mackerel scad?* for MLS; and *What is the biological rest period for mackerel scad?* for BRP), whereby knowledge of the policy interventions was defined by accurate responses. We found a considerable discrepancy between subjective and objective knowledge of both policy interventions (Table 4); that is, over 60% of respondents claimed to know about the MLS or BRP, but just over 30% gave correct answers; and the Pearson correlation coefficient between subjective and objective measures was only 0.5. The discrepancy was greatest among fishers. In their case, in particular, the level of objective knowledge about the policy interventions was very low. Fewer than 2% of the fishers gave correct answers to questions on both MLS and BRP. Given the importance of their role in maintaining the sustainability of fisheries and the fact that they are the main target group of the policy interventions, this finding is puzzling.

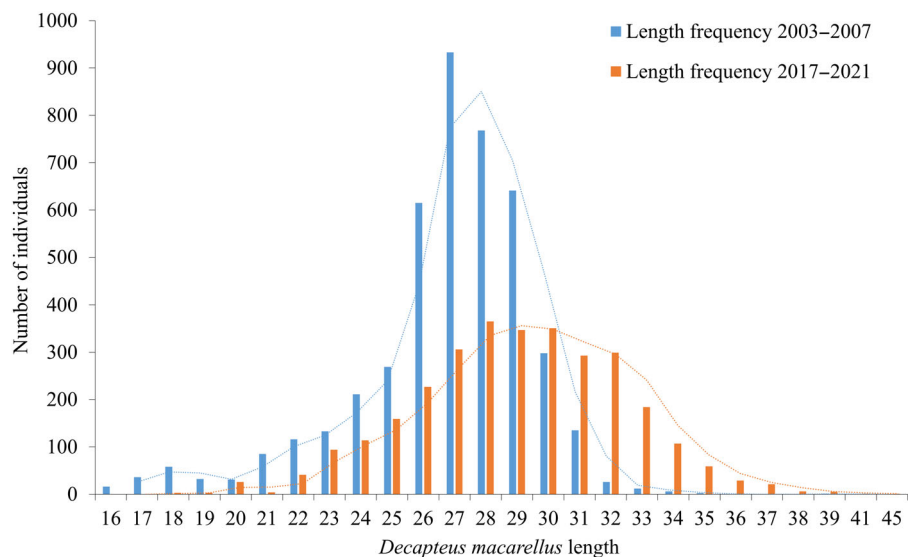


FIGURE 1 Length frequency of mackerel scad specimens during periods before (2003–2007) and after (2017–2021) the policy interventions.

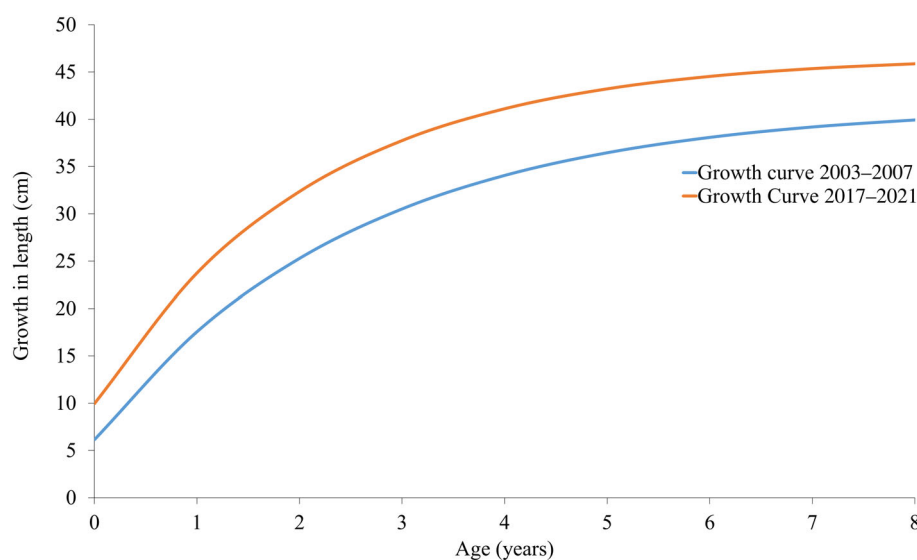


FIGURE 2 Von Bertalanffy growth curve of mackerel scad specimens during periods before (2003–2007) and after (2017–2021) policy interventions.

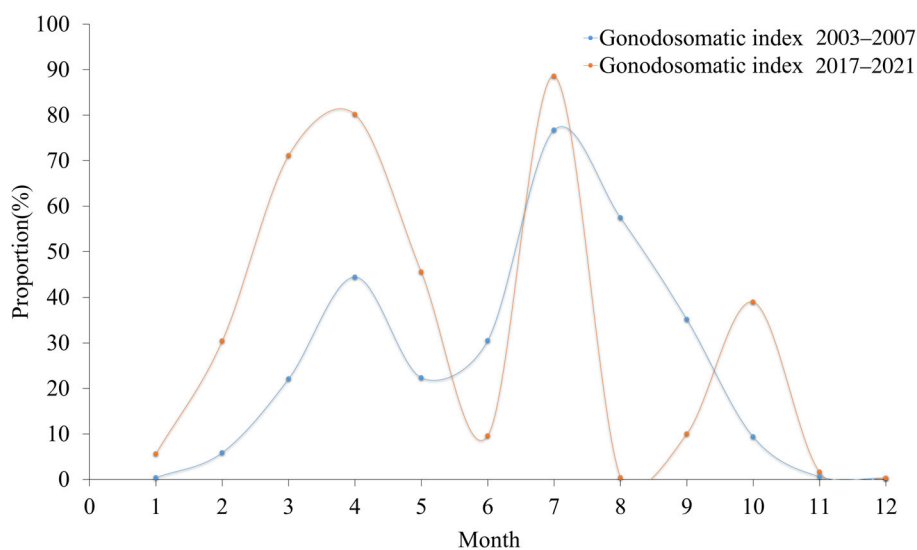


FIGURE 3 Gonadosomatic index of mackerel scad specimens during periods before (2003–2007) and after (2017–2021) the policy interventions.

FIGURE 4 Mature ogive of mackerel scad specimens during periods before (2003–2007) and after (2017–2021) policy interventions.

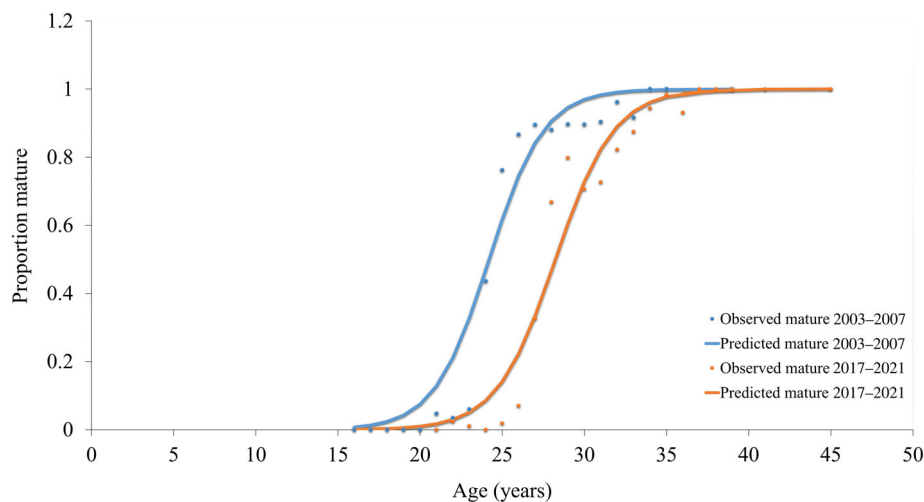


TABLE 3 Sociodemographic characteristics of the fishery stakeholders in São Vicente Island, who responded to the survey about fisheries policies implemented in Cabo Verde.

	All respondents	Fishers	Fishery institutions	Academia	NGOs	Consumers
Share of respondents (%)	100	33.71	26.68	10.86	7.43	21.14
Gender (%)						
Female	50.86	50.85	42.55	63.16	30.77	62.16
Male	49.14	49.15	57.45	36.84	69.23	37.84
Education (%)						
None	6.86	18.64	-	-	-	2.70
Primary school certificate	21.71	61.02	-	-	-	5.41
High school certificate	9.14	13.56	-	-	23.08	13.51
Bachelor's degree	37.14	-	53.19	52.63	46.15	64.86
Master's degree	14.29	-	27.66	21.05	30.77	10.81
Doctorate	5.14	-	8.51	26.32	-	-
Professional training	5.71	6.78	10.64	-	-	2.70
Years of experience (%)						
<5	33.14	5.08	12.77	52.63	61.54	83.78
6–15	29.14	27.12	46.81	31.58	23.08	10.81
16–25	15.43	15.25	25.53	15.79	15.38	2.70
26–35	11.43	22.03	12.77	-	-	2.70
36–45	5.71	15.25	2.13	-	-	-
46–55	5.14	15.25	-	-	-	-
No. of observations	175	59	47	19	13	37

Based on our results, having several years of professional experience is associated with a better understanding of policy interventions. Respondents with 6–35 years of experience are more likely to provide correct answers than those with less than 5 years of experience. However, respondents with more than 35 years of experience were less likely to give accurate answers. These effects mostly disappear, however, when controlling for occupational groups, that is, fishers, fishery institutions, academia, NGOs, and consumers, suggesting that individuals with certain characteristics self-select into occupations.

With regard to the perception of the status of mackerel stocks, fishery institutions, NGOs, and academia deemed the fisheries policies effective, as prior to policy interventions in 2008, the stocks were depleting (Table 5). However, fishers held the opposing belief that the stock was increasing, as, prior to the intervention, harvesting mackerel scad was comparatively effortless. Consumers lacked awareness of the trend. The policies received a good rating in general. With regard to its observed impacts, it was generally believed that there was a reduction in landing catch of mackerel scad, while its landing size and

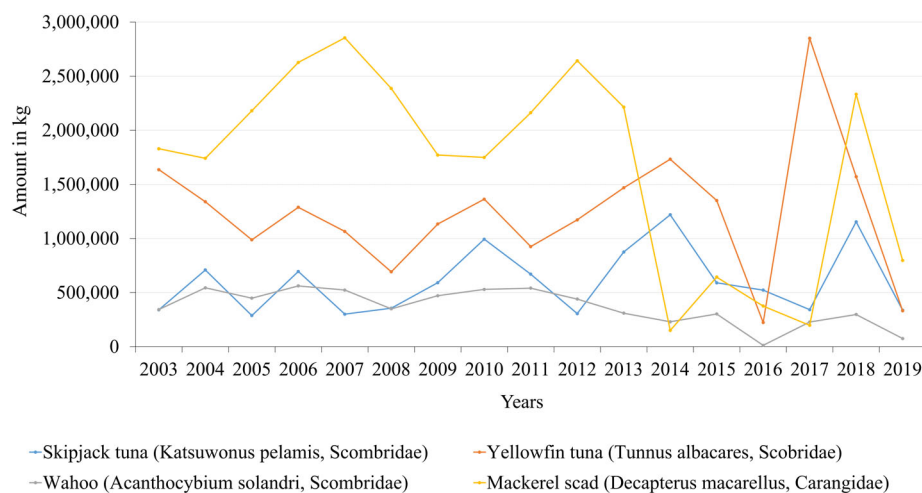


FIGURE 5 Landing catches from high trophic level tuna species and lower trophic mackerel scad between 2000 and 2019 (Data Source: IMar).

TABLE 4 Subjective and objective assessment of fishery stakeholders in São Vicente Island knowledge about fisheries policies implemented in Cabo Verde.

	All respondents	Fishers	Fishery institutions	Academia	NGOs	Consumers
Knowledge: Minimum landing size (%)						
Subjective	66.29	74.58	93.62	47.37	92.31	18.92
Objective	36	15.25	70.21	47.37	53.85	13.51
Knowledge: Biological rest period						
Subjective	64.57	79.66	80.85	68.42	61.54	18.92
Objective	33.14	18.64	61.70	47.37	46.15	8.11
Knowledge: Minimum landing size and biological rest period						
Subjective	51.43	59.32	76.60	36.84	61.54	10.81
Objective	20	1.69	48.94	26.32	38.42	2.70
No. of observations	175	59	47	19	13	37

TABLE 5 Perceptions about the policy interventions of fishery stakeholders in São Vicente Island knowledge about fisheries policies implemented in Cabo Verde before and after policies implementation.

	All respondents	Fishers	Fishery institutions	Academia	NGOs	Consumers
Perceptions about changes in mackerel scad (<i>Decapterus macarellus</i>) stock (%)						
Before the policy intervention						
Decreasing	53.71	28.81	85.11	68.42	69.23	40.54
Increasing	24.00	61.02	4.26	-	7.69	8.11
No changes	3.43	5.08	4.26	-	7.69	-
Don't know	18.86	5.08	6.38	31.58	15.38	51.35
After the policy intervention						
Decreasing	41.14	76.27	31.91	10.53	30.77	16.22
Increasing	29.14	18.64	46.81	31.58	23.08	24.32
No changes	5.14	-	10.64	10.53	7.69	2.70
Don't know	24.57	5.08	10.64	47.37	38.46	56.76
Overall opinion about the policy intervention (%)						
Good	72.57	72.88	85.11	68.42	53.85	64.86
Bad	11.43	18.64	4.26	15.79	15.38	5.41
No opinion	16.00	8.47	10.64	15.79	30.77	29.73
No. of observations	175	59	47	19	13	37

economic value increased. Likewise, when the respondents were asked about desired reforms to mackerel scad policies, there was a unanimous agreement on the necessity for improved monitoring and surveillance, and an adjustment of the MLS and a reduction in the BRP. Additional effective policies could involve awareness raising, limiting fishing to local vessels, conducting ecological studies of the natural habits of wild species, and investigating predator–prey relationships.

4 | DISCUSSION

Cabo Verde has policy instruments aimed at ensuring sustainable fisheries. However, a decadal (2009–2019) analysis of governance performance based on regulatory quality, an indicator that gauges the standard of policy formulation and implementation, as well as the perceived credibility of the government's commitment to such policies (Kaufmann et al., 2010), suggests a deteriorating regulatory quality (Osemwegie et al., 2022). In this study, there are divergent stakeholder views regarding the effectiveness of mackerel scad management policies, bringing to the fore some unanswered questions within the fisher community, such as “where are the mackerel scads hiding?” and “why have their numbers plummeted in marine catches?” Mackerel scad landing catch between 1992 and 2020 indicates that it is decreasing. Comparing the current results of the Nansen survey of Cabo Verdean waters with those from 2011, the 2021 survey (FAO, 2021) demonstrates a considerable decrease in both pelagic and demersal species. During the 2011 Nansen survey, mackerel scad, round scad (*Decapterus punctatus* Cuvier, 1829), and *Decapterus rhonchus* (Geoffroy Saint-Hilaire, 1817) (false scad) accounted for the main pelagic species identified. However, in the most recent survey of 2021, only a small number of round scad were caught, and none of the other two species were documented (FAO, 2021). The maximum sustainable yield (MSY) of mackerel scad has been estimated at 2400 t/year, suggesting a decline in the carrying capacity of its waters, which may be attributed to ocean warming (da Luz & Vieira, 2020).

Climate change is already affecting some of the main small pelagic fish, with potential implications for tropical nations (Barange et al., 2018). Each species has its own preferred optimal temperature, habitat, and diet (Morgan et al., 2010); changes in these conditions can directly affect fish survival, growth, and reproduction. Climate change impacts marine ecosystems by altering ocean physicochemical conditions, including SSTs, that affect primary productivity, species distribution, and food web structure (Cheung et al., 2014). In large marine ecosystems (LMEs), fisheries catch is influenced by changes in the primary productivity (Chassot et al., 2010). In addition, direct exploitation, that is, overfishing of top predators, can disrupt marine food chains (Cheung et al., 2010). Trophic dynamic modeling suggests that the predation mortality of mackerel scad, a forage fish species, decreased in response to a decrease in tuna abundance (World Bank, 2008). Data covering the pre-policy intervention period (between 2001 and 2012) showed that catch per unit of effort (CPUE)

for mackerel scad for artisanal purse seiner fleets and industrial fleets peaked in 2005 and 2006, respectively (Vieira, 2018). The CPUE decreased subsequently, indicating that mackerel scad was no longer a target (Vieira, 2018). Moreover, the increase in catches of big-eye scad and picarel could be attributable to the overexploitation of mackerel scad, enabling competing species to proliferate (World Bank, 2008). In tropical regions, most of the fish reproduction occurs during the rainy season that marks an increase in temperature, photoperiod, and water flow and serves as the stimulus for migration and spawning. Therefore, reproduction failure over several years has the potential to reduce wild fish stocks (Rizzo & Bazzoli, 2020).

4.1 | Mackerel scad stock state before and after the implementation of the fishery management policies

The average landing catch size and length frequency of mackerel scad increased after the policy intervention. Similar results were obtained by Santos (2018) where the FL was bigger (16–39 cm FL for the years 2005–2018) in comparison to Almada (1997b) assessment (23–27 cm FL for the years 1988–1996). This indicates that the implemented policies, MLS and BRP, have significantly impacted mackerel scad landing size. In these circumstances, mackerel scad has more time to grow before they are harvested. According to Beverton and Holt (1956), the growth of mackerel scad can be affected by several biotic or abiotic factors, such as food availability, population density, habitat modifications, and sea temperature (Beverton & Holt, 1956). Other biotic and abiotic factors include the prevailing conditions during the capture, the time of the year, their life cycle and sex (i.e., juvenile, male or female, and whether they were reproducing), the habitat they were occupying (whether it is favorable or not for their needs, and if they are currently migrating), and their gonad maturity (whether their reproductive organs are matured). Additionally, their diet, the time since their last meal, and their health condition can play a role in their sizes.

Costa et al. (2020) reported that females had an L_{50} of 24.1 cm FL, and males had an L_{50} of 26.6 cm FL between 2012 and 2018. Past studies show that the L_{50} is highly variable; using the same data, mackerel scad has had an L_{50} of 22.9 cm FL (Santos, 2018) and 20.30 cm FL (Vieira, 2018) in different time period. Therefore, there is a need for regular monitoring and amendment of the legal harvesting size to ensure sustainability. As with previous studies (Almada, 1997b; Santos, 2018; Vieira, 2018), two GSI peaks were identified (in April and July) for 2003/2007 and a considerably weaker peak in 2017/2021. The reproductive period with the highest peak is designated as the BRP. The emergence of a third peak during the post-policy intervention period suggests increasing spawning periodicity, which might imply favorable environmental conditions or increased predators' catch in the area.

Comparing the results, introduction of the fishery policies resulted in a reduction in the pressure on mackerel scad fishing. This has had a positive influence on its population, as they are reproducing

TABLE 6 Sociodemographic characteristics and knowledge about the policy instruments.

	Probit	
	Marginal effects	
	(1)	(2)
Dependant variable: knowledge about MLS or BRP		
Female	0.057	0.077
	0.421	0.258
Age (base category: 21–30)		
31–40	0.049	−0.037
	0.638	0.724
41–50	0.033	−0.037
	0.799	0.767
51–60	0.210	0.167
	0.144	0.289
61–70	0.094	0.052
	0.665	0.803
Education (base category: none)		
Primary school certificate	0.113	0.098
	0.354	0.496
High school certificate	0.091	0.038
	0.566	0.841
Bachelor's degree	0.469***	0.351
	0.000	0.102
Master's degree	0.441***	0.311
	0.002	0.166
Doctorate degree	0.432**	0.245
	0.029	0.359
Professional training	0.499***	0.416*
	0.008	0.051
Years of fishing experience (base category: <5)		
6–15	0.334***	0.145
	0.001	0.217
16–25	0.228	0.042
	0.089*	0.771
26–35	0.298*	0.133
	0.051	0.427
36–45	0.050	−0.090
	0.818	0.679
46–55	−0.205	−0.310*
	0.255	0.099
Occupation (baseline: fishers)		
Fishery institutions		0.164
		0.448
Academia		0.106
		0.652
NGOs		0.144

(Continues)

TABLE 6 (Continued)

	Probit	
	Marginal effects	
	(1)	(2)
		0.515
Consumers		−0.299*
		0.078
No. of observations	175	175

Abbreviations: BRP, biological rest period; MLS, minimum landing size; NGO, nongovernmental organization.

at a later stage and growing larger than before. Likewise, as suggested by Vieira (2018), the rise in sea temperature may also be contributing to these changes in the growth and first maturation length of mackerel scad, as Almada (1997b) reported annual average SST of 24.1°C with a maximum of 27 cm FL, and Vieira (2018) reported a relatively higher SST of 24.7°C with a maximum of 39 cm FL. Relating the L_{50} of mackerel scad across different regions, it was observed that habitat conditions greatly affect the L_{50} . In the waters of Japan, the mackerel scad L_{50} is 25.8 cm FL (Shiraishy et al., 2010), whereas in Hawaiian waters, the L_{50} is 24.5 cm (Clarke & Privitera, 1995).

4.2 | Stakeholders' perception of mackerel scad fishing management policies

Fishers collaborate more in fishery research because they are more experienced and understand the importance of sustainable fisheries. According to Ramos (1999), in the past, it was difficult to involve them in sustainable fishing studies. The different views on the state of the mackerel scad stock highlight the need for in-depth ecological research to complement local perceptions. However, fishers who were directly involved in wild fish production had results that were closer to reality. Fishers and those involved in fisheries resource management had a better understanding of the different policy instruments. The correlation between sociodemographic characteristics and objectively measured knowledge of the policy interventions showed that respondents with formal education, that is, professional training or university degree (bachelor, master, or doctorate), have a higher awareness of the policy interventions (Table 6). Therefore, there is a need for public awareness raising and wider dissemination of fishery policies. Currently, the NGO Biosfera I has implemented a project to raise awareness about fishery policies, targeting some of the fish species that are under these legislations. The implemented MLS, as well as the target species, is posted on the rear side of public buses. In addition, Gonzalez et al. (2020) consider that for Cabo Verde, policy implementation gaps in the fisheries sector are exacerbated by the lack of fiscalization policies, training of fishers, poverty among fishers, and the lack of evaluation and monitoring.

Despite legislation and measures put in place, mackerel scad catches continue to decrease. Changes in large-scale climate patterns due to both anthropogenic and natural causes can affect marine organisms (Geraldi et al., 2019). Island states are more exposed to effects of climate change (Gonzalez et al., 2020). This notwithstanding, Link et al. (2010) pointed out that commercial fishing had a greater impact than climatic change on soft bottom and pelagic marine ecosystems within the Northern Hemisphere. The harvesting of large apex predators can affect the ecology, and the SWOT analysis showed that this, on the one hand, affects all distribution, and, on the contrary, increases reliance on fish from the bottom of the marine food webs, such as the mackerel scad in other words, “fishing down marine food webs” (Pauly & Palomares, 2005). For example, the ratio of observed biomass to the biomass that would provide maximum sustainable yield (B/BMSY) of estimated mackerel scad stocks was estimated at 0.41 in Cabo Verde waters (Palomares et al., 2020), which is indicative of fully/overfished stocks and out of safe biological limits. Other countries in the Canary current large marine ecosystems (CCLME) where mackerels are grossly overfished are Mauritania, Senegal, Gambia, Guinea-Bissau, Guinea, and Sierra Leone (Palomares et al., 2020). However, mackerel scad is recorded in landing catches on Canary Islands, and, other mackerel species, *Decapterus tabl* (Berry, 1968), are increasing in Cabo Verde landing catches, as reported by the former National Director for Fisheries and Agriculture, Dr. Albertino Martins (Sapo Videos, 2018). In some instances, illegal, unreported, and unregulated (IUU) fishing practiced by international vessels could be contributing to mackerel scad decline, because fiscalization is not enforced in Cabo Verde due to the financial and logistical constraints. In addition to implementing and revising legislations and policy measures, new and updated socioecological studies and assessments should be conducted to improve fisheries management. For example, coastal zones, recognized to be potential nursery areas due to their topographic and oceanographic conditions (MAAP, 2004), are vulnerable to anthropogenic pressures like sand mining and sedimentation, which cause changes in these ecosystems National Institute of Fisheries Development (INDP, 2011b). Protecting these nursery areas can lead to a reduction in the mortality of mackerel eggs and juveniles, consequently enhancing stock recovery.

To build the resilience of fishers, capacity building is necessary to guide them to explore other less-known benthic species such as scorpionfishes, shrimps, and crabs (Macremar and Mariscomac, 2018). To facilitate this process, social inclusion is essential to strengthen collaboration between fisheries stakeholders, as there is a weak collaboration between fishers, scientists, and fisheries research/management institutions (Gonzalez et al., 2020).

5 | CONCLUSION

This study examined the political, ecological, and social impacts of the marine fisheries management policies of Cabo Verde. The findings showed that Cabo Verde has implemented effective policies and measures to promote sustainable fishing. Despite facing

constraints on management and resources, the government has developed policies that align with relevant international conventions aimed at sustaining a sustainable blue economy. However, there are some gaps and shortcomings that need to be addressed to improve the effective implementation of fishery policies. First, a thorough understanding of ecological resources, their abundance and distribution, and the ecosystems they inhabit is needed to develop policies that are better suited to the exploitation of marine resources. This is a significant knowledge gap that needs to be addressed, as knowledge of the habitats of mackerel scad, as well as associated biotic and abiotic factors in Cabo Verdean waters, is limited. For instance, it is unclear why the mackerel scad from landing catches are growing larger. This increase in their sizes could be due to the MLS and BRP policies, changes in their natural habitat, or a combination of both factors. Furthermore, the cause for their decline in marine landing catches, as stated by the fishers, remains indeterminate.

Overfishing large pelagic fish and climate change impacts on mackerel habitat might also be contributing to mackerel scad decline. Other potential causes of their decline could be behavioral changes due to changes in their habitat, as well as negative impacts on their life stages, including spawning and early stages. However, it is challenging to make these assumptions, given the lack of studies supporting these hypotheses.

The study concludes that most fishery stakeholders are aware of the fisheries policies in a subjective way; however, some fishers do not understand the importance of such measures for stock recovery; they perceive that such policies cause the mackerel scad to disappear from the fishing ground and thus reduce their incomes. There is, therefore, the need for a shift to evidence-based fishery policies and the inclusion of relevant stakeholders in the policy-making process.

AUTHOR CONTRIBUTIONS

Katelen Delgado conceptualized the research idea, data curation, formal analyses, writing - original draft preparation, and writing - review and editing. Isimemen Osemwegie contributed to conceptualization, writing - original draft preparation, and writing - review and editing and supervision. Anibal Delgado Medina reviewed the methodology, provision of laboratory materials, and writing - review and editing. Alciany Nascimento da Luz reviewed the methodology, data curation, and writing - review and editing. Zaneta Kubik performed formal analysis, reviewed the methodology, and writing - review and editing. Essetchi Paul Kouamelan validated and reviewed the methodology, analytical tools, writing - original draft preparation, and writing - review and editing, and supervision.

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REFERENCES

- Almada, E. O. (1997a). Life history of scad mackerel *Decapterus macarellus* Cuvier, 1833, in the waters off the Cape Verde Islands. IMar Report.
- Almada, E. O. (1997b). Life history of scad mackerel *Decapterus macarellus* Cuvier, 1833, in the waters off the Cabo Verde islands. Master's Dissertation, Haskoli Islands, Reykjavik, Iceland. p. 66.
- Barange, M., Bahri, T., Beveridge, M. C. M., Cochrane, K. L., Funge-Smith, S., & Poulain, F. (Eds.). (2018). *Impacts of climate change on fisheries and aquaculture: Synthesis of current knowledge, adaptation, and mitigation options*. FAO fisheries and aquaculture technical paper No. 627 (p. 628). Rome.
- Belhabib, D., Campredon, P., Lazar, N., Sumaila, U. R., Baye, B. C., Abou Kane, E., & Pauly, D. (2016). Best for pleasure, not for business: Evaluating recreational marine fisheries in West Africa using unconventional sources of data. *Palgrave Communications*, 2(1), 1–10.
- Belhabib, D., Sumaila, U. R., & Pauly, D. (2015). Feeding the poor: Contribution of West African fisheries to employment and food security. *Ocean & Coastal Management*, 111, 72–81.
- Béné, C., Barange, M., Subasinghe, R., Pinstrup-Andersen, P., Merino, G., Hemre, G. I., & Williams, M. (2015). Feeding 9 billion by 2050—putting fish back on the menu. *Food Security*, 7, 261–274.
- Beverton, R. J. H., & Holt, S. J. (1956). On the dynamics of exploited fish populations. U.K. Ministry of Agriculture and fisheries. *Fishery Investigation*, 19(2), 533.
- Cadima, E. L. (2000). Manual de Avaliação de Recursos Pesqueiros. FAO Doc Téc Pescas, 393, 162.
- Canyon, D., Allen, E., Long, M., & Brown, C. (2021). Policy recommendations for combating overfishing and fisheries crime. N2628-Canyon-et-al-policy-recommendations-for-combatting-overfish.pdf. ([jstor.org](https://doi.org/10.3389/fmars.2019.00030))
- Chassot, E., Bonhommeau, S., Dulvy, N., Mélin, F., Watson, R., Gascuel, D., & Le Pape, O. (2010). Global marine primary production constraints fisheries catch. *Ecology Letters*, 13(4), 495–505.
- Cheung, W., Ota, Y., & Swartz, W. (2014). *Predicting future oceans climate change*. Oceans and Fisheries. The Nipon Foundation-University of British Columbia NEREUS PROGRAM.
- Cheung, W. W., Lam, V. W., Sarmiento, J. L., Kearney, K., Watson, R., Zaller, D., & Pauly, D. (2010). Large-scale redistribution of maximum fisheries catches potential in the global ocean under climate change. *Global Change Biology*, 16, 24–35.
- Clarke, T. A., & Privitera, L. (1995). Reproductive biology of two Hawaiian pelagic carangid fishes, the bigeye Scad, *Selar Crumenophthalmus*, and the round Scad, *Decapterus macarellus*. *Bulletin of Marine Science-Miami*, 56, 33–47.
- Costa, M. P. V., Cruz, D. R. S., Monteiro, L. S., Evora, K. S. M., & Cardoso, L. G. (2020). Reproductive biology of the mackerel scad *Decapterus macarellus* from Cabo Verde and the implications for its fishery management. *African Journal of Marine Science*, 42(1), 35–42.
- Cuvier, G., & Valenciennes, A. (1833). Histoire naturelle des poissons. Tome neuvième. Suite du livre neuvième. Des Scomberoides. v. 9: i-xxix + 3 pp. + 1-512, P246-279.
- da Luz, A., & Vieira, N. (2020). Stock assessment of blackspot picarel (*Spi-cara melanurus*) and mackerel scad (*Decapterus macarellus*) in Cape Verde waters, Northwest Africa. In M. L. D. Palomares, M. Khal-fallah, J. Woroniak, & D. Pauly D (Eds.), *Assessments of marine fisheries resources in West Africa with emphasis on small pelagics* (Vol. 28, pp. 5–17). Fisheries Centre Research Report.
- DeAlteris, J. (2012). *Assessment of stock status and intensity of exploitation for the cavala preta, chicharro and dobrada, small pelagic fishery resources in the waters surrounding Cabo Verde*. Intermediate report, DeAlteris associates Inc. Document submitted to: West Africa Regional Fisheries Project-Cabo Verde (WARFP-CV), Praia, Santiago, Cabo Verde.
- DNA. (2015a). V Relatório Nacional Sobre o Estado Da Biodiversidade em Cabo Verde. Direção Nacional do Ambiente Report, Mindelo, Cabo Verde p. 92.
- DNA. (2015b). *Estratégia Nacional e Plano de Ação para a Conservação da Biodiversidade 2015–2030*. Direção Geral do Ambiente Report (p. 100). Praia.
- Duarte, M. C., & Romeiras, M. M. (2009). Cabo Verde Islands. In R. G. Gillespie & D. A. Clague (Eds.), *Encyclopedias of islands. Encyclopaedias of the natural world*, No.2 (pp. 143–148). University of California Press.
- FAO. (1995). Code of conduct for responsible fisheries. Food & Agriculture Origination, Rome 5p.
- FAO. (2020). *The state of world fisheries and aquaculture 2020*. Sustainability in action.
- FAO. (2021). Nansen marine ecosystem survey in Cabo Verde: Key highlights. Available from Nansen marine ecosystem survey in Cabo Verde: Key highlights | EAF Nansen | Food and Agriculture Organization of the United Nations. ([fao.org](https://www.fao.org)).
- Gayani, F. C., & Sparre, P. (2005). FAO-ICLARM stock assessment tools II: User's guide (No. 8). Food & Agriculture Org.
- Gerald, N. R., Kellison, G. T., & Bache, N. M. (2019). Climate indices, water temperature, and fishing predict broad scale variation in fishes on temperate reefs. *Frontiers in Marine Science*, 6, 30. <https://doi.org/10.3389/fmars.2019.00030>
- Gonzalez, J. A., Monteiro, C. A., Correia, S., Lopes, E., Almeida, N., Martins, A., Gaztanaga, I., Gonzalez-Lorenzo, G., Arenas-Ruiz, R., Tejera, G., & Lorenzo, J. M. (2020). Current and emerging small-scale fisheries and target specimens in Cabo Verde, with recommendations for pilot actions favouring sustainable development. *Cybio*, 44(4), 355–371.
- González, J. A., & Tariche. (2009). *Um olhar sobre a biodiversidade marinha e bases para a sua gestão sustentável* (p. 176). Potenciais recursos pesqueiros de profundidade de Cabo Verde/ una mirada sobre la biodiversidad marina y bases para su gestión sostenible. Recursos pesqueros potenciales de profundidad de Cabo Verde. Presidencia del Gobierno de Canarias/Fundación Universitaria de Las Palmas.
- Haddon, M. (2001). *Modelling and quantitative methods in fisheries* (p. 406). Chapman & Hall/CRC Press.
- Hay, J. E. (2013). Small Island developing states: Coastal systems, global change, and sustainability. *Sustainability Science*, 8, 309–326.
- Hijmans, R. J., Cameron, S. E., Parra, J. L., Jones, P. G., & Jarvis, A. (2005). Very high-resolution interpolated climate surfaces for global land areas. *International Journal of Climatology*, 25, 1965–1978. Worldclim database. Available on: <http://www.worldclim.org/bioclim>.
- Hilborn, R., Amoroso, R. O., Anderson, C. M., Baum, J. K., Branch, T. A., Costello, C., De Moor, C. L., Faraj, A., Hively, D., Jensen, O. P., & Kurota, H. (2020). Effective fisheries management instrumental in improving fish stock status. *Proceedings of the National Academy of Sciences*, 117(4), 2218–2224.
- IPBES. (2018). In E. Archer, L. Dziba, K. J. Mulongoy, M. A. Maoela, & M. Walters (Eds.), *The IPBES regional assessment report on biodiversity and ecosystem services for Africa* (p. 492). Secretariat of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services.

- Kaufmann, D., Kraay, A., & Mastruzzi, M. (2010). *The worldwide governance indicators: Methodology and analytical issues* [online]. World Bank. Available at: <https://openknowledge.worldbank.org/handle/10986/3913>
- Kawarazuka, N., & Béné, C. (2011). The potential role of small fish specimens in improving micronutrient deficiencies in developing countries: Building evidence. *Public Health Nutrition*, 14(11), 1927–1938.
- Kelemen, E., Pataki, G., Konstantinou, Z., Varumo, L., Paloniemi, R., Pereira, T. R., Sousa-Pinto, I., Vandewalle, M., & Young, J. (2021). Networks at the science-policy-interface: Challenges, opportunities and the viability of the 'network-of-networks' approach. *Environmental Science & Policy*, 123, 91–98.
- King, M. (1995). Population dynamics. In M. King (Ed.), *Fisheries biology, assessment and management* London: Fishing new books, Oxford (pp. 193–287). BlackWell Publishing.
- Link, J. S., Yemane, D., Shannon, L. J., Coll, M., Shin, Y.-J., & Hill, L. (2010). Relating marine ecosystem indicators to fishing and environmental drivers: An elucidation of contrasting responses. *ICES Journal of Marine Science*, 67(4), 787–795.
- Macremar & Mariscomac. (2018). Informe técnico sobre las experiencias de pesca experimental de prospección de cantarero (peces escorpénidos) en Cabo Verde. S. Vicente, Cabo Verde: Proyecto MARISCOMAC Report. Mindelo, Cabo Verde.
- Martins, P. A. R., & Knúttsson, Ö. (2002). Management of a new Fleet for expansion of fisheries in the Cape Verde Islands: A feasibility study. *UNU fisheries training Programme*, United Nations University, PO box (1390), Skulagata, 4, 120.
- McNaughton, D. B. (2008). A biological and social examination of opelu (*Decapterus spp.*) fisheries in the west Hawaii, Hawaii islands. A thesis submitted to the Tropical Conservation Biology and Environmental Science Graduate Program at the University of Hawaii at Hilo in Partial Fulfilment of the Requirements for the Degree of Master of Science.
- Ministério do Ambiente, Agricultura e Pesca (MAAP). (2004). Projecto de Cooperação Técnica na Área de Investigação Científica sobre Recursos Haliêuticos na Zona Económica Exclusiva da República de Cabo Verde, 2004. *MAAP Report Cabo Verde*, 25–39.
- Monnereau, I., Mahon, R., McConney, P., Nurse, L., Turner, R., & Vallès, H. (2021). Fisheries sector vulnerabilities to climate change in Small Island developing states. In *Small Island Developing States: Vulnerability and Resilience Under Climate Change* (pp. 233–255). Springer.
- Monteiro, F., Fortes, A., Ferreira, V., Pereira Essoh, A., Gomes, I., Correia, A. M., & Romeiras, M. M. (2020). Current status and trends in Cabo Verde agriculture. *Agronomy*, 10(1), 74.
- Morgan, M. J., Rideout, R. M., & Colbourne, E. B. (2010). *Impact of environmental temperature on Atlantic cod Gadus morhua energy allocation to growth, condition and reproduction*. (p. 11). Marine Ecology Progress Series.
- National Institute of Fisheries Development. (2011a). *Análise das Instituições e Políticas de Pesca em Cabo Verde* (p. 127). APPECCAO-Cabo Verde Report.
- National Institute of Fisheries Development. (2011b). *Trabalhos apresentados na VIII Reunião Ordinária do Conselho Científico. National Institute of fisheries development report* (p. 202). Mindelo.
- National Institute of Fisheries Development (INDP). (2004). *Projeto de Cooperação técnica na Área de Investigação Científica sobre Recursos Haliêuticos na Zona Económica Exclusiva da República de Cabo Verde. National Institute of Fisheries Development Report* (p. 81). Mindelo.
- National Institute of Fisheries Development (INDP). (2007). *Trabalhos apresentados na VII Reunião Ordinária do Conselho Científico. National Institute of Fisheries Development Report* (p. 229). Mindelo.
- National Institute of Fisheries Development (INDP). (2012). *Relatório de Principais Resultados do Censo Geral da Frota de Pesca Artesanal e Industrial/Semi-Industrial ANO de 2011*.
- Organisation for Economic Co-operation and Development (OECD). (2016). *Reference guide on ex-post evaluation of competition agencies' enforcement decisions* (p. 100). Organisation for Economic Co-operation and Development report.
- Osemwegie, I., Delgado, K. D. C., Arimiyaw, A. W., Kanneh, A. B., Todota, C. T., Faye, A., & Akinyemi, F. O. (2022). Diagnostic analysis of the canary current system of West Africa: The need for a paradigm shift to proactive natural resource management. *Ocean and Coastal Research*, 69, p.e21043.
- Palomares, M. L. D., Khalfallah, M., Woroniak, J., Pauly, D., & Us, S. A. (2020). Assessment of 14 species of small pelagic fish caught along the coast of northwest African countries. *Fisheries Centre Research Reports*, 28, 4.
- PANA II. (2004). *Segundo Plano de Ação Nacional para o Ambiente – Plano de Gestão dos Recursos da Pesca*, Ministério do Ambiente, Agricultura e Pescas e Gabinete de Estudos e Planeamento Report (p. 218). Praia.
- Pauly, D., & Moreau, J. (1997). *Méthodes pour l'Évaluation des Ressources Halieutiques* (p. 288). Cepaduès-Éditions.
- Pauly, D., & Palomares, M. L. (2005). Fishing down marine food web: It is far more pervasive than we thought. *Bulletin of Marine Science*, 76(2), 197–212.
- PEDS. (2017). *Plano Estratégico de Desenvolvimento Sustentável 2017/2021* (p. 212). Ministério das Finanças Report.
- R Core Team (2013). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. ISBN 3-900051-07-0, URL: <http://www.R-project.org/>.
- Ramos, V. (1999). A pescaria dos pequenos pelágicos em Cabo Verde. Instituto Nacional para o Desenvolvimento da Pesca (INDP), Reunion du conseil Scientifique, 8-9 Décembre 1999, p. 12.
- República de Cabo Verde. Resolução No. 1/2016, de 28 de Janeiro de. (2016). *Prorroga a validade do Plano Executivo Bianual de Gestão dos Recursos da Pesca para 31 de Março de 2016. Boletim Oficial da República de Cabo Verde de 2016, Imprensa Nacional de Cabo Verde, SA. I Série No. 5* (p. 187). Conselho de Ministros.
- República de Cabo Verde. Resolução No. 10/2009, de 4 Maio de. (2009). *Aprova o Plano Executivo Bianual de Gestão dos Recursos das Pescas (PGRP) para o período de 1 de Janeiro de 2009 a 31 de Dezembro de 2010. Boletim Oficial da República de Cabo Verde de 2009, Imprensa Nacional de Cabo Verde, SA. I Série No. 18* (pp. 1560–1565). Conselho de Ministros.
- República de Cabo Verde. Resolução No. 11/2007, de 2 de Abril de. (2007). *Aprova o Plano de Gestão dos Recursos da Pesca para o período de 1 de Março de 2007 a 31 de Dezembro de 2008. Boletim Oficial da República de Cabo Verde de 2007, Imprensa Nacional de Cabo Verde, SA. I Série No. 18* (pp. 174–178). Conselho de Ministros.
- República de Cabo Verde. Resolução No. 29/2016, de 16 de Março de. (2016). *Aprova o Plano Executivo Bianual de Gestão dos Recursos da Pesca 2016–2017. Boletim Oficial da República de Cabo Verde de 2016, Imprensa Nacional de Cabo Verde, SA. I Série No.16* (pp. 474–482). Conselho de Ministros.
- República de Cabo Verde. Resolução No. 3/2005, de 21 de Fevereiro de. (2005). *Aprova o Plano de Gestão dos Recursos da Pesca. Boletim Oficial da República de Cabo Verde de 2005, Imprensa Nacional de Cabo Verde, SA. I Série No. 32. Conselho de Ministros, Praia, Cabo Verde*, pp. 240–244.
- República de Cabo Verde. Resolução No. 56/2014, de 28 de Julho de. (2014). *Aprova o Plano Executivo Bianual de Gestão dos Recursos das Pescas para o período de 1 de Janeiro de 2014 a 31 de Dezembro de 2015. Boletim Oficial da República de Cabo Verde de 2014, Imprensa Nacional de Cabo Verde, SA. I Série No. 45* (pp. 1560–1565). Conselho de Ministros.
- República de Cabo Verde. Resolução No. 63/2021, de 18 de junho de. (2021). *Aprova a suspensão dos períodos de defeso para a cavala e chicharro para o ano 2021. Boletim Oficial da República de Cabo Verde de 2021, Imprensa Nacional de Cabo Verde, SA. I Série No. 61* (pp. 1865–1866).
- República de Cabo Verde. Resolução No. 8/2018, de 1 de Fevereiro de. (2018). *Aprova o Plano Executivo Bianual de Gestão dos Recursos da*

- Pesca 2016–2017. *Boletim Oficial da República de Cabo Verde de 2018, Imprensa Nacional de Cabo Verde, SA. I Série No. 16* (p. 75). Conselho de Ministros.
- República de Cabo Verde. Resolução No. 93/2020, de 9 de Julho de. (2020). *Aprova o Plano Executivo Anual de Gestão dos Recursos da Pesca para o ano de 2020. Boletim Oficial da República de Cabo Verde de 2020, Imprensa Nacional de Cabo Verde, SA. I Série No. 81* (pp. 1620–1626). Conselho de Ministros.
- Rizzo, E., & Bazzoli, N. (2020). *Reproduction and embryogenesis. In biology and physiology of freshwater neotropical fish* (pp. 287–313). Academic Press.
- Robinson, S. A., Bouton, E., Dolan, M., Meakem, A., Messer, A., Lefond, I., & Roberts, J. T. (2022). A new framework for rapidly assessing national adaptation policies: An application to small Island developing states in the Atlantic and Indian oceans. *Regional Environmental Change*, 22(2), 37.
- Sambe, B., Lymer, B. L., Caramelo, A. M., & Tandstad, M. (2011). *Reversing the degradation of the canary current large marine ecosystem* (p. 19). In ICES Annual Science Conference.
- Santos, M. G. (2018). *Dinâmica populacional da cavala preta (Decapterus macarellus) na região norte de Cabo Verde: diagnóstico do período de 2005-2018. Bachelor Thesis. Universidade de Cabo verde.*
- Shiraishy, T., Tanaka, H., Ohshimo, S., Ishida, H., & Morinaga, N. (2010). Age, growth and reproduction of two specimens of scad *Decapterus macrossoma* and mackerel scad (*Decapterus macarellus*, Carangidae) in the waters off southern Kyushu. *Japan Agricultural Research Quarterly*, 44(2), 197–206.
- Sparre, P., & Venema, S. C. (1997). In S. C. Venema (Ed.), *Introdução à avaliação de mananciais de peixes tropicais in Sparre, P. FAO Documento Técnico sobre as Pescas.*
- Tariche, O., Correia, M., Cruz, E., & e Fonseca, A. (2011). *O defeso da cavala preta: Causas, impactos e alternativas. In: Trabalhos apresentados na VIII reunião ordinária do conselho científico, Mindelo, 6 a 7 Novembro de 2008 Instituto Nacional Desenvolvimento das Pescas* (pp. 158–175). National Institute of Fisheries Development (INDP).
- Tariche, O., & Martins, A. (2006). *Dinâmica populacional e avaliação do estado de exploração dos principais recursos haliêuticos de Cabo Verde.* In National Institute of Fisheries Development (INDP), Report, Mindelo São Vicente.
- Teh, L. S. L., Teh, L. C. L., & Sumaila, U. R. (2013). A global estimate of the number of coral reef fishers. *PLoS One*, 8(6), e65397.
- Sapo Videos, 2018. Cavala de rabo vermelho aparece nas águas de Cabo Verde, por causa das mudanças climáticas. Disponível em: Cavala de rabo vermelho aparece nas águas de Cabo Verde, por causa das mudanças—SAPO Vídeos. Consulted on November 2021.
- Vieira, N. (2018). *Stock assessment and the influence of environmental parameters on the distribution of mackerel scad (Decapterus macarellus) in Cabo Verde waters.* United Nations University Fisheries Training Program. <http://www.unuftp.is/static/fellows/document/Nuno18prf.pdf>.
- Von Bertalanffy, L. (1938). A quantitative theory of organic growth (inquiries on growth laws. II). *Human Biology*, 10(2), 181–213.
- World Bank, (2008). Assessment, fisheries sector strategy. "Cape Verde." Available at: <https://documents1.worldbank.org/curated/en/377811468006264103/pdf/AAA210ESW0Whit10strategy0assessment.pdf>
- World Bank, (2021). Cabo Verde Data. Available at: <https://data.worldbank.org/country/cabo-verde>.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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